

Abdullah Khaled

Programming Lead & Chief Technical Advisor, ITKAN FRC 9128 — FIRST® Leadership Award Semi-Finalist (2026)

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Education

Wakeland High School, High School Diploma – Frisco, TX Aug 2023 – May 2027

- Class rank: top 5 of ~560; unweighted GPA: 4.0; 18 AP courses by graduation (completed and planned), typically 4–6 AP/year in STEM

Frisco ISD Career and Technical Education Center, Project Lead The Way (Engineering) – Frisco, TX Aug 2023 – May 2027

- PLTW Nationally Distinguished Engineering Program — IED, POE, Digital Electronics, EDD
- [Engineering portfolio](#)

Extracurricular Activities

Programming Lead, ITKAN Robotics — FIRST Robotics Competition (FRC 9128) – Plano, TX Sept 2024 – Feb 2026

- Led robot **Programming** as primary Java/WPILib developer for a ~50-member team: owned ~80% of the codebase (2024-12–2025-06), GitHub workflows, command-based autonomous and teleop
- Built AprilTag pose estimation + auto-align, swerve odometry fusion, and a field-oriented driver UI; integrated OpenCV + SolvePnP with Kalman fusion; shipped autonomous pathing/targeting (state machines, Bezier, PID)—~50% higher autonomous points/match across 12 matches after vision integration
- Designed and fabricated an Onshape camera mount for reliable AprilTag detection across field distances; implemented driver-input buffering, custom motor logging classes (AdvantageScope), and match logging/simulation for limited build-season access
- **Mentored** rookie FTC teams 26357 and 28931 (2024–Present); supported ~30 students (6 programmers) through competition, including ~25 hrs/week at peak FTC build season
- **Led** multiple FLL workshops (7 students/session) for refugee students using the **conceived** OARobotics curriculum ([OARobotics](#)); students completed block-code autonomous maze navigation by session end

Chief Technical Advisor, ITKAN Robotics — FIRST Robotics Competition (FRC 9128) – Plano, TX July 2025 – present

- **Lead** software strategy for ITKAN's multi-team FRC program (~20 programmers, ~100-member organization): delegate work, standardize codebase layout, align autonomy/teleop with mechanical and **Electrical** integration under build-season deadlines
- Managed autonomy/teleop reliability (>90% target) with layered failsafes; preserved match performance after a near-failure camera disconnect through contingency planning
- Improved mechanism cycle time (~2 s to <1 s), shooter accuracy (~70% to ~95%), throughput (~10 to ~20 balls/s, lab-measured); completed autonomous routines without error in ~8–10 pre-competition practice runs
- Presented vision and autonomous work to engineers at corporate sponsors **Toyota** and **Texas Instruments** represented the team at nationwide conferences and community/university showcases (incl. U.S. Senator Chris Van Hollen event)
- **Led** summer FRC programming training (~10 students) from fundamentals through programming a competition robot from scratch; **conceived** and built adaptive training site [Mantik](#) (written lessons, code, video); YouTube tutorial playlist; Discord support for peer learning
- **Conceived** a lightweight team task system to coordinate build-season work and reduce downtime between matches; cooperative leadership: adaptive curriculum by student mastery, follow-ups tying lessons to live robot code

Founder and President, Wolverine Robotics — Wakeland High School (FTC 33791) – Frisco, TX Aug 2024 – present

- Founded school FTC program after difficulty finding a school-sponsored FIRST team; **led** club operations and connected members with local FIRST opportunities while teaching Java, CAD, and competition skills
- Built competition-ready robot in <2 weeks; UIL State Championship 3rd place; grew membership from 0 to 50+ (year one); ~\$6k+ resourcing secured (school, NHS, parents)
- Delivered ~6 workshops (10–15 attendees) and monthly demonstrations; **Outreach**: promoted STEM via school administration presentations and engineering notebook

Lead Instructor & Peer Program Coordinator, Ma'ruf Dallas Refugee Organization (Project Taleem) – Dallas, TX Sept 2023 – present

- Co-lead tutoring for K–6 refugees (~2 hr/week school year); multilingual, age-banded groups; recruited volunteers from five high schools via robotics networks
- Measured ~15-point average IXL SmartScore gain (Nov–Feb; n=10); monthly robotics enrichment using OARobotics for inclusive, hands-on STEM access

Vice President, Mu Alpha Theta Math Honor Society, Wakeland High School – Frisco, TX Aug 2024 – present

- **Led** annual middle school math competition: 100+ participants, 3 schools, 35+ volunteers; expanded Math Lab peer tutoring (>20% member participation)

Co-Founder, Secretary, Muslim Student Association (MSA), Wakeland High School – Frisco, TX Aug 2024 – present

- Co-founded club; secured sponsor and prayer space; created outreach media for student community building

Work Experience

Math and Reading Instructor, Gideon Math and Reading Center – Frisco, TX June 2024 – present

- Instructs K–11 (through Algebra/Geometry) in guided sessions (~4–12 students/day); uses performance data to personalize plans and pacing

Math and Reading Teaching Assistant (TA), Gideon Math and Reading Center – Frisco, TX Jan 2022 – May 2024

- Graded multi-grade work; improved grading efficiency ~50%

STEM Projects

OralVision (Oral Disease Detector) - Conrad Challenge; Diamond Challenge Oct 2025 – present

- On-device imaging + privacy-preserving ML pipeline; 3D-printed device (Onshape); Conrad Innovator Award; Diamond Semifinalist

WatchFall (Senior Fall Monitor) – Samsung Solve for Tomorrow Sept 2025 – present

- Raspberry Pi Pico + TFLite fall detection; hybrid dataset to reduce false positives

Honors & Awards

- (2026) FIRST® Tech Challenge: FIRST® Leadership Award Semi-Finalist (Team 28931)
- (2025) FRC 9128: Event Winner (FiT Victoria, 25 teams); Autonomous Award (FiT Plano, 31 teams); Creativity Award (FiT Amarillo, 36 teams)
- (2026) FTC 33791: Event Winner; Control Award (FiT-North Dallas Semi-Regional, 25 teams); Control Award (FiT-North U-League)
- (2026) OralVision: Conrad Challenge Innovator Award; Diamond Challenge Semifinalist
- (2025) FTC 26357: Finalist Alliance Captain (FiT-North McKinney Qualifier, 19 teams)
- (2025) College Board: AP Scholar with Distinction; National Recognition Program — School Recognition Award